



AJKD Atlas of Renal Pathology: Malakoplakia

Mark A. Lusco, MD,¹ Agnes B. Fogo, MD,¹ Behzad Najafian, MD,² and Charles E. Alpers, MD²

Clinical and Pathologic Features

Patients with malakoplakia usually present with fever, flank and/or loin pain, and a history of *E. coli* urinary tract infection. Malakoplakia may occur at any age, but most commonly presents in the fifth decade of life with a predilection for women over men (4:1). Patients may be immunosuppressed, with underlying conditions such as AIDS, diabetes, alcohol abuse, or another immunocompromised state. Bilateral kidney involvement occurs in less than half of cases and usually presents with acute kidney injury. Prognosis has markedly improved with use of fluoroquinolone drugs, with >90% patient survival.

Light microscopy: Sheets of macrophages, forming mass-like lesion, with granular eosinophilic cytoplasm and characteristic inclusions of periodic acid–Schiff–positive Michaelis–Gutmann bodies, which also stain for calcium and iron.

Immunofluorescence microscopy: Noncontributory.

Electron microscopy: The macrophage intracytoplasmic Michaelis–Gutmann bodies have crystalline central core with peripheral lamellar rings of mineral deposits.

Etiology/Pathogenesis

Macrophage inability to release lysosomal enzymes leads to decreased degradation of bacteria, most commonly *E. coli*. This defect in macrophage intracytoplasmic bactericidal function results in partially digested bacterial products that form a nidus for calcium and iron minerals.

Differential Diagnosis

Renal cell carcinoma can be distinguished from malakoplakia by positive cytokeratin staining in the absence of macrophage stains (for CD163 and CD68). Xanthogranulomatous pyelonephritis has lipid-laden

foamy macrophages without intracytoplasmic Michaelis–Gutmann bodies. Megalocytic interstitial nephritis has macrophages with granular eosinophilic cytoplasm, also without intracytoplasmic Michaelis–Gutmann bodies. Megalocytic interstitial nephritis is also closely linked to *E. coli* urinary tract infection and may represent an earlier stage of malakoplakia.

Key Diagnostic Features

- Sheets of macrophages with granular eosinophilic cytoplasm
- Cytoplasmic Michaelis–Gutmann bodies in macrophages

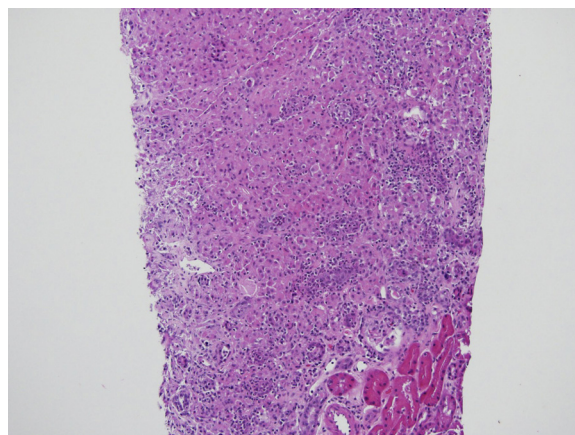


Figure 1. Malakoplakia with mass-like sheets of macrophages with granular eosinophilic cytoplasm and scattered neutrophils within the interstitium and tubules (hematoxylin and eosin stain).

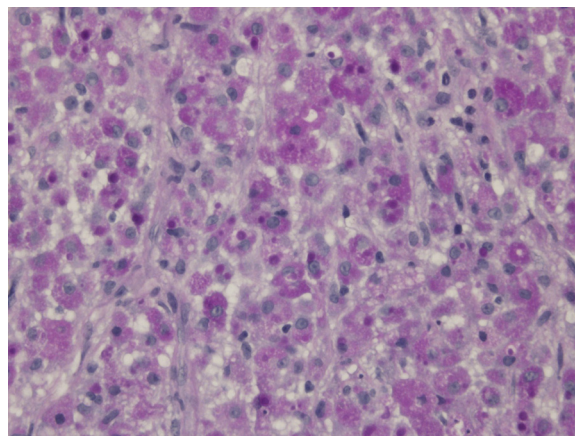


Figure 2. Malakoplakia with sheets of macrophages with granular cytoplasm and intracytoplasmic Michaelis–Gutmann bodies (periodic acid–Schiff stain).

From the ¹Department of Pathology, Microbiology and Immunology, Vanderbilt University, Nashville, TN; and ²Department of Pathology, University of Washington, Seattle, WA.

Support: None.

Financial Disclosure: The authors declare that they have no relevant financial interests.

Address correspondence to agnes.fogo@vanderbilt.edu

Am J Kidney Dis. 68(4):e27–e28.

© 2016 by the National Kidney Foundation, Inc.

0272-6386

<http://dx.doi.org/10.1053/j.ajkd.2016.08.006>

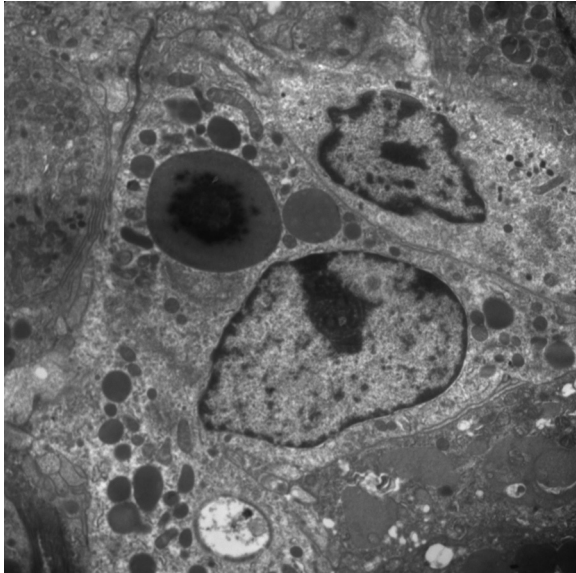


Figure 3. Malakoplakia with macrophage with intracytoplasmic Michaelis-Gutmann body composed of crystalline central core with peripheral mineral deposits (electron microscopy).

